



KAMBALA

Student Number: _____

Task 4
September 2016

Preliminary HSC Mathematics

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black pen
- Board-approved calculators may be used
- Answer Questions 1 – 10 on the multiple choice answer sheet
- Answer Questions 11 – 14 on the paper provided
- A reference sheet is provided with this paper
- In Questions 11 – 14 show relevant mathematical reasoning and calculations

Total marks – 70

Section I **10 marks**

- Attempt Questions 1 - 10
- Allow about 15 minutes for this section

Section II **60 marks**

- Attempt Questions 11 - 14
- Allow about 1 hour and 45 minutes for this section

Section I

10 Marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section

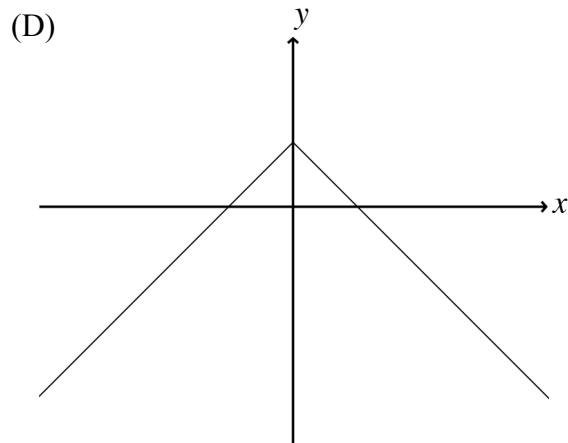
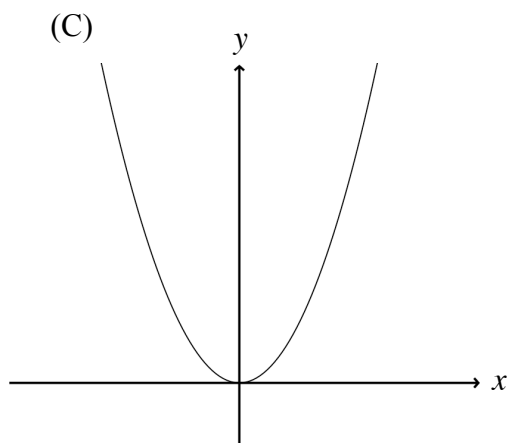
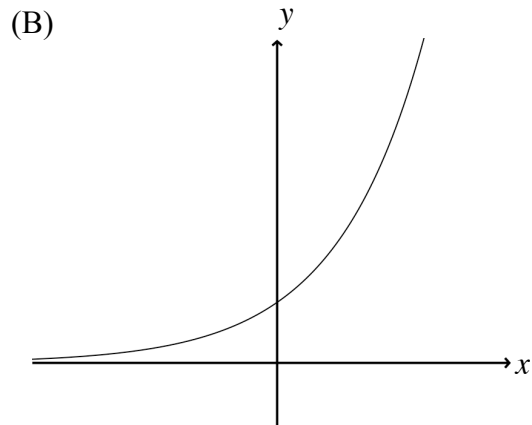
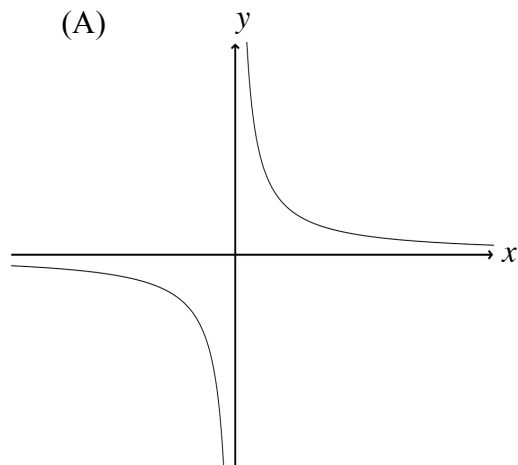
Use the answer sheet for Questions 1 – 10.

- 1 Evaluate $\sqrt{\frac{13.24 \times 3.7}{0.45}}$, giving your answer correct to four significant figures.
- (A) 10.43 (B) 10.433 (C) 10.434 (D) 10.4337

- 2 The exact value of $\sin 60^\circ \tan 30^\circ$ is:

- (A) $\frac{3}{2}$ (B) $\frac{\sqrt{3}}{2}$ (C) $\frac{1}{2\sqrt{3}}$ (D) $\frac{1}{2}$

- 3 Which of the following graphs represent a function that is neither odd nor even?

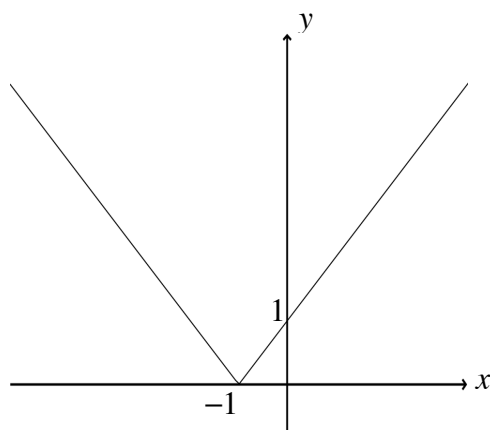


4 Let $f(x) = \sqrt{x-5}$. Which of the following is the domain of $f(x)$?

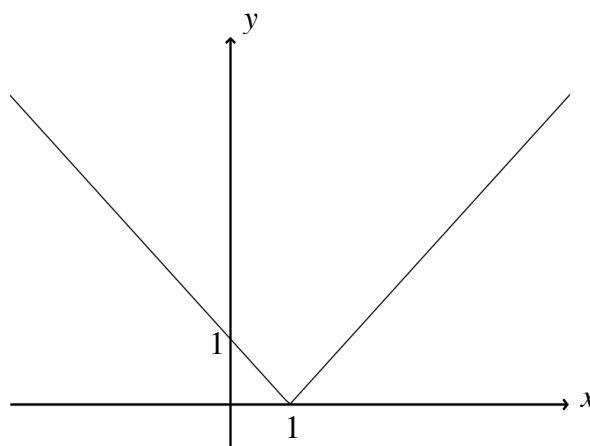
- (A) All real x (B) $x \geq 5$ (C) $x \leq 5$ (D) $-5 \leq x \leq 5$

5 Which graph shows $y = |x+1|$?

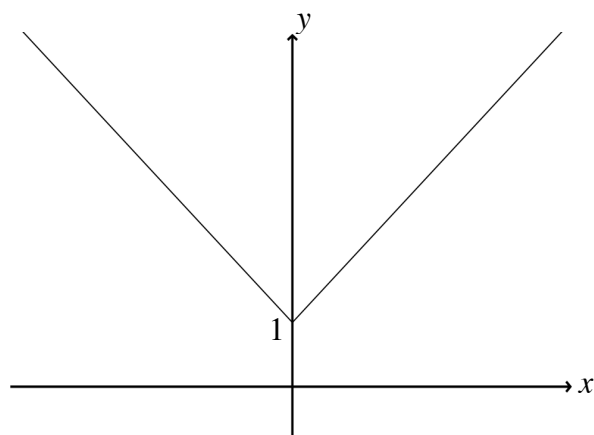
(A)



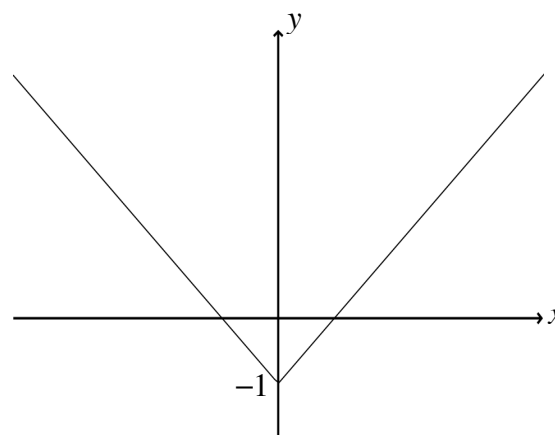
(B)



(C)



(D)



6 The solutions to the inequality $x^2 + 3x - 4 \leq 0$ are represented by:

- (A) $x \leq -4, x \geq 1$ (B) $x \leq -1, x \geq 4$ (C) $-4 \leq x \leq 1$ (D) $-1 \leq x \leq 4$

7 The locus of a point that moves so that its distance from the y -axis is always three times its distance from the x -axis is:

- (A) $y = \pm 3x$ (B) $y = \pm \frac{x}{3}$ (C) $y = \pm 3x^2$ (D) $y = \pm \frac{x^2}{3}$

8 If $\operatorname{cosec} \theta = -\frac{5}{3}$ and $\cos \theta > 0$, then the value of $\cot \theta$ is:

- (A) $-\frac{3}{4}$ (B) $\frac{3}{4}$ (C) $-\frac{4}{3}$ (D) $\frac{4}{3}$

9 In order to find the gradient of $f(x) = x^2 - 3x + 2$ using first principles, which one of the following expressions would, need to be simplified?

(A) $\lim_{h \rightarrow 0} \frac{(x+h)^2 - 3(x+h) + 2 - x^2 - 3x + 2}{h}$

(B) $\lim_{h \rightarrow 0} \frac{(x+h)^2 - 3(x+h) + 2 + x^2 - 3x + 2}{h}$

(C) $\lim_{h \rightarrow 0} \frac{(x+h)^2 - 3(x+h) + 2 - x^2 + 3x - 2}{h}$

(D) $\lim_{h \rightarrow 0} \frac{(x+h)^2 - 3(x+h) + 2 - x^2 - 3x - 2}{h}$

10 The expression $\frac{x^2 - 5x + 6}{4 - x^2}$ is equivalent to:

- (A) $\frac{6-5x}{4}$ (B) $\frac{x-3}{2+x}$ (C) $\frac{x-3}{x-2}$ (D) $\frac{3-x}{2+x}$

End of Section I

Section II**60 Marks****Attempt Questions 11 - 14****Allow about 1 hour 45 minutes for this section**

Answer each question on the writing paper provided. Start each question on a new page.

In Questions 11 - 14, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks)

- (a) The two shorter sides of a right-angled triangle have sides of lengths $2 + \sqrt{3}$ and $2 - \sqrt{3}$. Find the exact length of the hypotenuse. 2
- (b) Find the value of x given that $\cos 30^\circ = \sin(x + 20)^\circ$. 1
- (c) Simplify $\frac{2^{x-1} \times 2^3}{2^{4-3x}}$. 2
- (d) Given that $f(x) = x^2 + 1$:
- (i) Evaluate $f(-5)$. 1
- (ii) Find the value(s) of x such that $f(x) = 5$. 2
- (e) Find the values of k for which $3x^2 - 4x + k = 0$ has no real roots. 2
- (f) If α and β are the roots of the quadratic equation $2x^2 - 7x + 12 = 0$, find the value of:
- (i) $\alpha + \beta$. 1
- (ii) $\alpha\beta$. 1
- (iii) $\alpha^2 + \beta^2$. 2
- (g) For what value(s) of x is the graph of $f(x) = 1 + \frac{1}{x-2}$ discontinuous? 1

Question 12 (15 marks) Start a new page.

(a) If $2x^2 + 3x - 5 \equiv A(x+1)^2 + B(x+1) + C$, find A , B and C . **3**

(b) A parabola has equation $y^2 = 8x$.

(i) Find the

(I) focal length. **1**

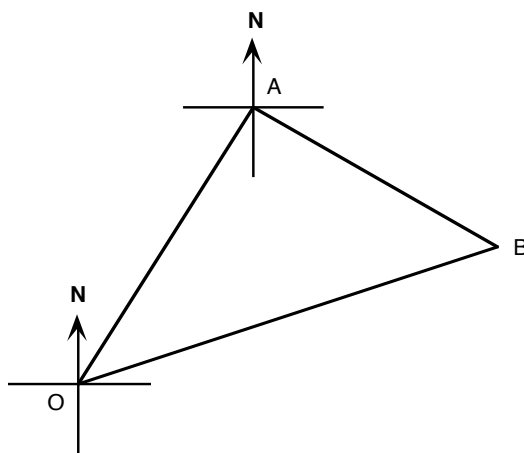
(II) coordinates of the vertex. **1**

(III) coordinates of the focus. **1**

(IV) equation of the directrix. **1**

(ii) Sketch the parabola indicating the above features. **1**

(c) A ship starts from O and sails 80 kilometres on a bearing of 035° to A . It then changes course and sails 55 kilometres on a bearing of 110° to B .



(i) Copy the diagram, marking on it the information supplied. **1**

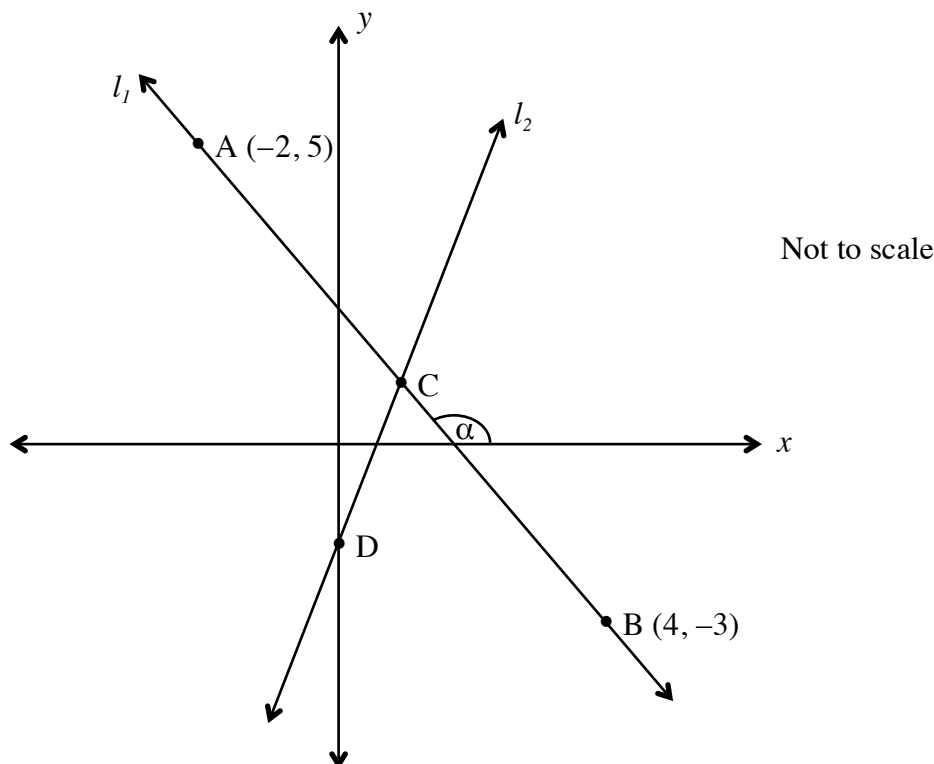
(ii) Show that $\angle OAB = 105^\circ$. **1**

(iii) Calculate the distance of B from O , correct to 1 decimal place. **2**

(d) Solve the equation $x^6 - 7x^3 - 8 = 0$ for x . **3**

Question 13 (15 marks) Start a new page.

- (a) Points $A(-2, 5)$ and $B(4, -3)$ lie on the line l_1 as shown in the diagram below.
 C is the point of intersection of lines l_1 and l_2 and is the midpoint of the interval AB .



- | | | |
|--------|--|---|
| (i) | Show that the gradient of line l_1 is $-\frac{4}{3}$. | 1 |
| (ii) | Hence, show that the equation of line l_1 is $4x + 3y - 7 = 0$. | 2 |
| (iii) | Find the size of α , the angle of inclination of line l_1 , to the nearest degree. | 1 |
| (iv) | Find the coordinates of C , the midpoint of AB . | 2 |
| (v) | The equation of line l_2 is $2x - y - 1 = 0$. Find the coordinates of D , the point where line l_2 crosses the y -axis. | 1 |
| (vi) | Find the length of AD , leaving your answer in simplest surd form. | 2 |
| (vii) | The line through AD has equation $3x + y + 1 = 0$. Find the shortest distance between the point C and this line. | 2 |
| (viii) | Hence, or otherwise, find the area of $\triangle ACD$. | 1 |

Question 13 continues on the next page

Question 13 (continued)

- (b) (i) A point $P(x, y)$ moves so that its distance from the point $A(1, 5)$ is equal to its distance from the point $B(4, -1)$. **2**

Show that the locus of P is a straight line with equation $2x - 4y + 3 = 0$.

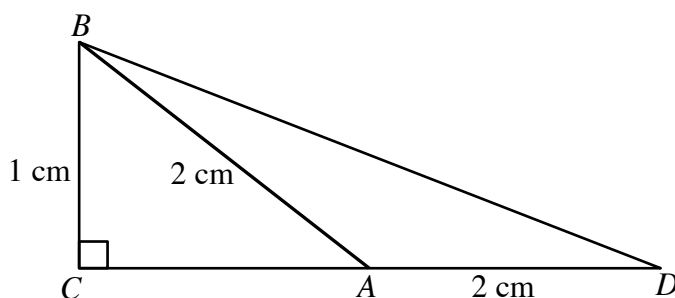
- (ii) Describe geometrically the locus of P . **1**

Question 14 (15 marks) Start a new page.

- (a) A function is defined by the rule:

$$f(x) = \begin{cases} x^2 + 4 & x \geq 0 \\ \frac{1}{x} & x < 0 \end{cases}$$

- (i) Find $f(k^2)$ 1
- (ii) Sketch the graph of $y = f(x)$ showing all its features. 2
- (b) Consider the equation $x^2 + (m-3)x + m = 0$. Find the values of m for which the equation has two real and different roots. 3
- (c) Solve the equation $2 \sin(\theta - 60)^\circ = -\sqrt{3}$ for $-180^\circ \leq \theta \leq 180^\circ$. 3
- (d) The locus of a point $P(x, y)$ is a parabola with equation $x^2 - 8x + 4y - 12 = 0$. Find the coordinates of the vertex of the parabola. 2
- (e) Triangle ABC is right-angled at C with $AB = 2$ cm and $BC = 1$ cm. D is a point on CA produced such that $AD = 2$ cm as shown in the diagram below.



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- (i) Show that $\angle ADB = 15^\circ$. 2
- (ii) Hence show that the exact value of $\tan 15^\circ = 2 - \sqrt{3}$. 2

End of Section II

Student Number: _____

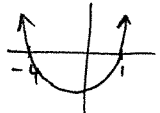
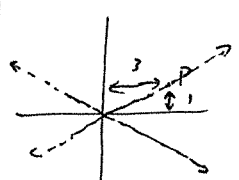
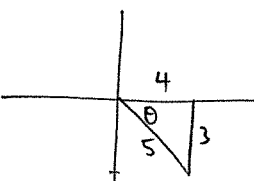
Mathematics

Task 4 Preliminary HSC Examination September 2016

Section I

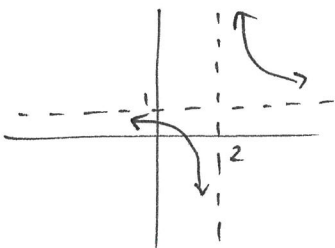
Multiple-Choice Answer Sheet *Circle your response*

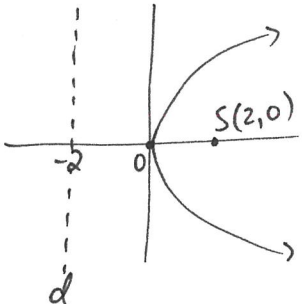
- | | | | | |
|-----|---|---|---|---|
| 1. | A | B | C | D |
| 2. | A | B | C | D |
| 3. | A | B | C | D |
| 4. | A | B | C | D |
| 5. | A | B | C | D |
| 6. | A | B | C | D |
| 7. | A | B | C | D |
| 8. | A | B | C | D |
| 9. | A | B | C | D |
| 10. | A | B | C | D |

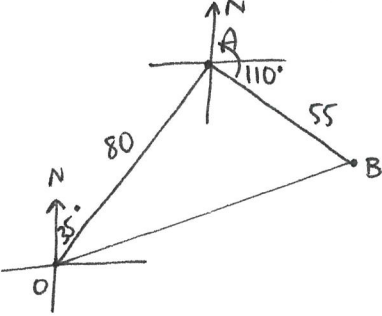
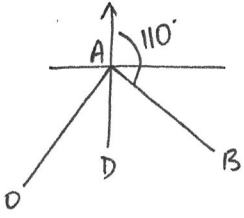
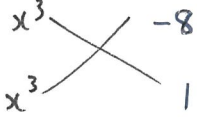
Qn	Solutions	Marks	Comments & Criteria
	Section 1 :		
1.	(A) 10.43	1	
2.	$\frac{\sqrt{3}}{2} \times \frac{1}{\sqrt{3}} = \frac{1}{2}$ (D)	1	
3.	(B)	1	
4.	$x - 5 \geq 0$ $x \geq 5$ (B)	1	
5.	(A)	1	
6.	$(x + 4)(x - 1) \leq 0$  $-4 \leq x \leq 1$ (C)	1	
7.	 $y = \pm \frac{1}{3}x$ (B)	1	
8.	$\sin \theta = -\frac{3}{5}$  $\tan \theta = -\frac{3}{4}$ $\cot \theta = -\frac{4}{3}$ (C)	1	

Qn	Solutions	Marks	Comments & Criteria
9.	$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ $\lim_{h \rightarrow 0} \frac{(x+h)^2 - 3(x+h) + 2 - (x^2 - 3x + 2)}{h}$ $\lim_{h \rightarrow 0} \frac{(x+h)^2 - 3(x+h) + 2 - x^2 + 3x - 2}{h}$ (c)	1	
10.	$\frac{(x-3)(x-2)}{(2-x)(2+x)}$ $\frac{(x-3)(\cancel{x-2})}{-1(\cancel{x-2})(2+x)}$ $\frac{(x-3)}{-1(2+x)}$ $-1 \times \frac{(x-3)}{2+x}$ $\frac{3-x}{2+x} \quad (d)$	1	

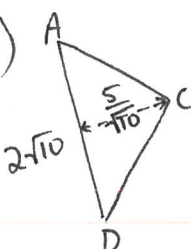
Qn	Solutions	Marks	Comments & Criteria
Q11	Section II:		
(a)	$c^2 = (2+\sqrt{3})^2 + (2-\sqrt{3})^2$ $c^2 = 4 + 4\sqrt{3} + 3 + 4 - 4\sqrt{3} + 3$ $c^2 = 14$ $c = \sqrt{14}$	1 for working 1 for exact area	
(b)	$\cos 30^\circ = \sin(90^\circ - 30^\circ)$ $\cos 30^\circ = \sin 60^\circ$ $\sin 60 = \sin(x + 20)$ $x = 40$	1 for answer	
(c)	$2(x-1) + 3 - (4-3x)$ $= 2^{4x-2}$	1 for working 1 for answer	
(d) i)	$f(-5) = (-5)^2 + 1$ $= 25 + 1$ $= 26$	1 for answer	
ii)	$5 = x^2 + 1$ $x^2 = 4$ $x = \pm 2$	1 for working 1 for answer	

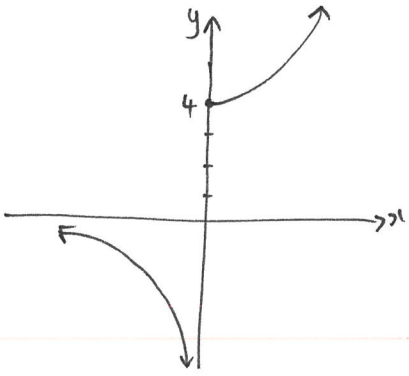
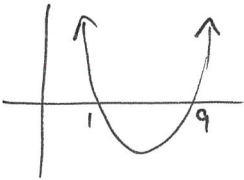
Qn	Solutions	Marks	Comments & Criteria
(e)	$b^2 - 4ac < 0$ $16 - 12k < 0$ $-12k < -16$ $k > \frac{4}{3}$	1 for working 1 for answer	
(f) i)	$\alpha + \beta = \frac{-b}{a}$ $= \frac{7}{2}$	1 for answer	
ii)	$\alpha\beta = \frac{c}{a}$ $= \frac{12}{2}$ $= 6$	1 for answer	
iii)	$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$ $= \left(\frac{7}{2}\right)^2 - 2(6)$ $= \frac{49}{4} - 12$ $= \frac{1}{4}$	1 for working 1 for answer	
(g)	$x = 2$ 	1 for answer (must be an equation)	

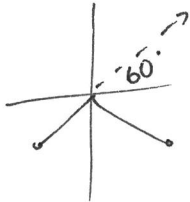
Qn	Solutions	Marks	Comments & Criteria
Q12 (a)	$A(x+1)^2 + B(x+1) + C$ $A(x^2 + 2x + 1) + Bx + B + C$ $Ax^2 + 2Ax + A + Bx + B + C$ $Ax^2 + (2A+B)x + (A+B+C)$ $\underline{A = 2}$ $2A + B = 3$ $4 + B = 3$ $\underline{B = -1}$ $A + B + C = -5$ $2 - 1 + C = -5$ $\underline{C = -6}$		
(b)i)	$y^2 = 8x$ (i) focal length: $4a = 8$ $a = 2$ (ii) vertex: $(0, 0)$ (iii) focus: $(2, 0)$ (iv) directrix: $x = -2$	1 for each A, B & C. ↳ if incorrect values, progress marks may be awarded. 1 1 1 1	
ii)		1	

Qn	Solutions	Marks	Comments & Criteria
(c) i)			1 for all information
ii)	 $\angle DAB = 180^\circ - 110^\circ = 70^\circ$ $\angle OAD = 35^\circ \text{ (alternate angles)}$ $\therefore \angle OAB = 70 + 35 = 105^\circ$	1	
iii)	$x^2 = 80^2 + 55^2 - 2 \times 80 \times 55 \times \cos 105$ $x^2 = 11702.6076$ $x = 108.2 \text{ km}$	1 for working 1 for answer	
(d)	$(x^3)^2 - 7x^3 - 8 = 0$  $(x^3 - 8)(x^3 + 1) = 0$ $x^3 = 8 \quad x^3 = -1$ $x = 2 \quad x = -1$	1 for factorisation 1 for each answer	

Qn	Solutions	Marks	Comments & Criteria
Q13 a)	i) $m = \frac{y_2 - y_1}{x_2 - x_1}$ $= \frac{-3 - 5}{4 + 2}$ $= \frac{-8}{6}$ $= -\frac{4}{3}$ ii) $y + 3 = -\frac{4}{3}(x - 4)$ $3y + 9 = -4x + 16$ $4x + 3y - 7 = 0$ iii) $m = \tan \theta$ $-\frac{4}{3} = \tan \theta$ $\theta = 53^\circ$ $\theta = 180 - 53 = 127^\circ$ iv) $M = \left(\frac{-2 + 4}{2}, \frac{5 - 3}{2} \right)$ $= (1, 1)$	1 for working to show gradient 1 for working 1 for answer 1 for answer 1 for working 1 for answer	
	v) $2x - y - 1 = 0$ let $x = 0$ $-y - 1 = 0$ $y = -1$ D: $(0, -1)$	1 for coordinates	

Qn	Solutions	Marks	Comments & Criteria
	vi) $d = \sqrt{(-2-0)^2 + (5+1)^2}$ $d = \sqrt{(-2)^2 + 6^2}$ $d = \sqrt{4+36}$ $d = \sqrt{40}$ $d = 2\sqrt{10}$ vii) $\frac{ ax_1 + by_1 + c }{\sqrt{a^2 + b^2}}$ $= \frac{ 3 + 1 + 1 }{\sqrt{9+1}}$ $= \frac{5}{\sqrt{10}}$ viii)  $A = \frac{1}{2}bh$ $= \frac{1}{2} \times 2\sqrt{10} \times \frac{5}{\sqrt{10}}$ $= 5 \text{ units}^2$	1 for working 1 for answer in simplest surd form 1 for working 1 for answer 1 for answer	

Qn	Solutions	Marks	Comments & Criteria
(b)	i) $PA^2 = PB^2$ $(x-1)^2 + (y-5)^2 = (x-4)^2 + (y+1)^2$ $x^2 - 2x + 1 + y^2 - 10y + 25 = x^2 - 8x + 16 + y^2 + 2y + 1$ $6x - 12y + 9 = 0$ $2x - 4y + 3 = 0$ ii) Perpendicular bisector of AB	1 for working 1 for answer 1 for appropriate geometric description	
Q14 a)	i) $f(k^2) = (k^2)^2 + 4$ $= k^4 + 4$ ii) 	1 for answer 1 for each part of the curve -1 for any errors	
(b)	$b^2 - 4ac > 0$ $(m-3)^2 - 4m > 0$ $m^2 - 6m + 9 - 4m > 0$ $m^2 - 10m + 9 > 0$ $(m-9)(m-1) > 0$ $m < 1, m > 9$ 	1 for substitution of Δ 1 for factorisation 1 for answer	

Qn	Solutions	Marks	Comments & Criteria
(c)	$2 \sin (\theta - 60^\circ) = -\sqrt{3}$ $\sin (\theta - 60^\circ) = \frac{-\sqrt{3}}{2}$ $-180^\circ \leq \theta \leq 180^\circ$ $-240^\circ \leq \theta - 60 \leq 120^\circ$ $-240^\circ \leq u \leq 120^\circ$ $\sin u = \frac{-\sqrt{3}}{2}$ $u = -60^\circ, -120^\circ$ $\text{since } x = u + 60$ $\theta = 0^\circ, -60^\circ$		1 for domain 1 for working with 'u' 1 for answers
(d)	$x^2 - 8x + 4y - 12 = 0$ $x^2 - 8x = -4y + 12$ $(x-4)^2 = -4y + 12 + 16$ $(x-4)^2 = -4y + 28$ $(x-4)^2 = -4(y-7)$ $V: (4, 7)$		1 for working 1 for vertex

Qn	Solutions	Marks	Comments & Criteria
(e) i)	$a^2 + b^2 = c^2$ $a^2 + 1 = 4$ $a^2 = 3$ $a = \sqrt{3}$ $\therefore CD = 2 + \sqrt{3}$ $\tan \theta = \frac{1}{2 + \sqrt{3}}$ $\theta = 15^\circ$		<u>Option 1:</u> 1 to get to CD 1 for showing $\theta = 15^\circ$
ii)	$\tan 15 = \frac{1}{2 + \sqrt{3}}$ $\frac{1}{2 + \sqrt{3}} \times \frac{2 - \sqrt{3}}{2 - \sqrt{3}} = \frac{2 - \sqrt{3}}{4 - 3}$ $= 2 - \sqrt{3}$ $\therefore \tan 15 = 2 - \sqrt{3}$		<u>Option 2:</u> 1 for $\angle CAB = 30^\circ$ 1 for recognition of isosceles & $\theta = 15^\circ$ 1 for tan ratio statement 1 for rationalising.